



3 (a) State the principle of moments.

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..... [1]

(b) A rigid uniform beam rests on a pivot at its centre, as shown in Fig. 3.1.

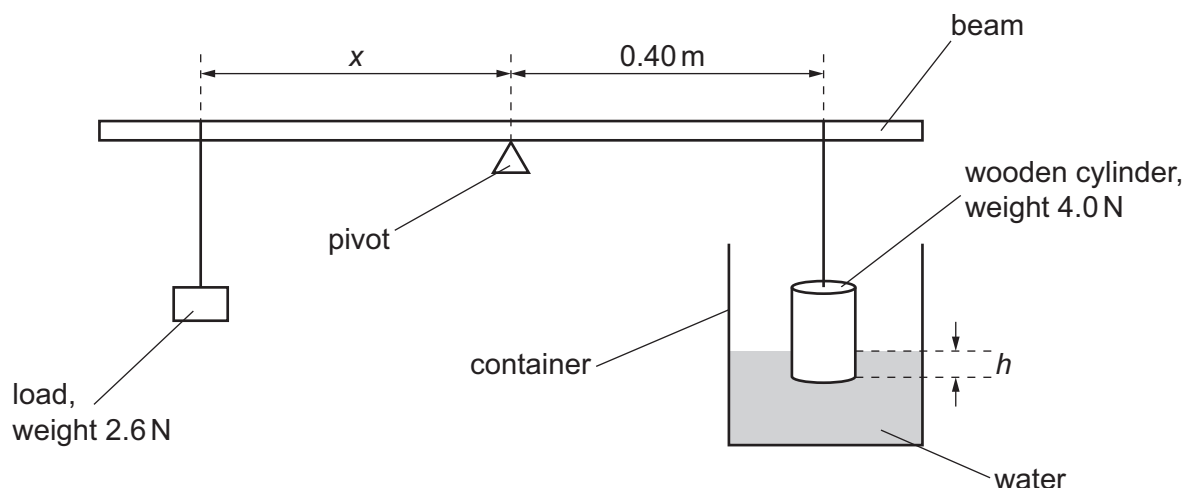


Fig. 3.1 (not to scale)

A load of weight 2.6 N is suspended from the beam at distance x from the pivot.

A wooden cylinder of weight 4.0 N is suspended from the beam at a distance of 0.40 m from the pivot on the opposite side of the pivot to the load. The cylinder rests in a container of water. The lower part of the cylinder is immersed in the water to depth h .

Initially, h is equal to 0.10 m and x is equal to 0.40 m. The system is in equilibrium.

- (i) Use the principle of moments to show that the upthrust U exerted by the water on the cylinder is 1.4 N.

[2]



- (ii) The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

Calculate the area A of the circular cross-section of the cylinder.

$$A = \dots\dots\dots \text{ m}^2 \text{ [3]}$$

- (c) More water is gradually added to the container in (b), so that depth h in Fig. 3.1 gradually increases. The length x is continuously adjusted so that the system remains in equilibrium.

On Fig. 3.2, sketch the variation of x with h . Use the space below for any working.

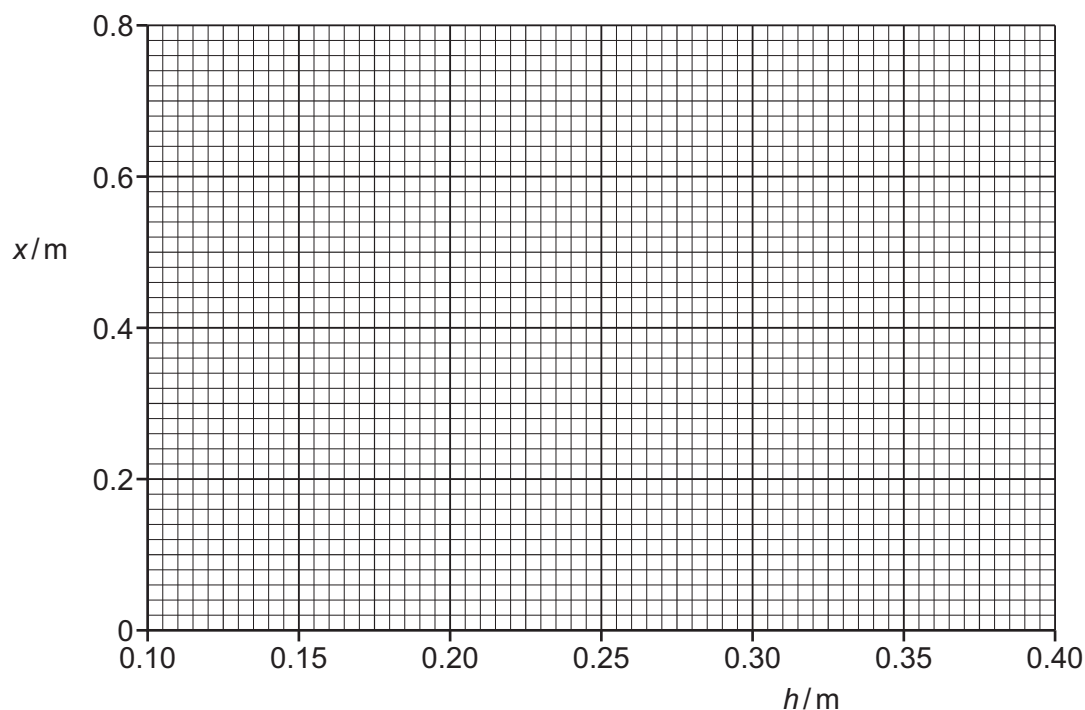


Fig. 3.2

[3]

[Total: 9]